

MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 5

Roman Isachenko

Moscow Institute of Physics and Technology
Yandex School of Data Analysis

Recap of Previous Lecture

Training

1. Sample $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$, $\epsilon \sim p(\epsilon)$.
2. Reparametrize $\mathbf{z} = \mathbf{g}_{\phi}(\mathbf{x}, \epsilon)$.
3. Compute the ELBO:

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) \approx \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})).$$

4. Update ϕ , θ via stochastic gradient ascent.

Sampling

1. Sample $\mathbf{z} \sim p(\mathbf{z}) = \mathcal{N}(0, \mathbf{I})$.
2. Sample $\mathbf{x} \sim p_{\theta}(\mathbf{x}|\mathbf{z})$.

Note: The encoder $q_{\phi}(\mathbf{z}|\mathbf{x})$ isn't needed during sampling.

Recap of Previous Lecture

Theorem

$$\frac{1}{n} \sum_{i=1}^n D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}_i) \parallel p(\mathbf{z})) = D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z}) \parallel p(\mathbf{z})) + \mathbb{I}_q[\mathbf{x}, \mathbf{z}].$$

Revisiting the ELBO

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \mathcal{L}_{\phi,\theta}(\mathbf{x}_i) &= \underbrace{\frac{1}{n} \sum_{i=1}^n \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x}_i)} \log p_{\theta}(\mathbf{x}_i|\mathbf{z})}_{\text{Reconstruction Loss}} \\ &\quad - \underbrace{\mathbb{I}_q[\mathbf{x}, \mathbf{z}]}_{\text{Mutual Information}} - \underbrace{D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z}) \parallel p(\mathbf{z}))}_{\text{Marginal KL}}. \end{aligned}$$

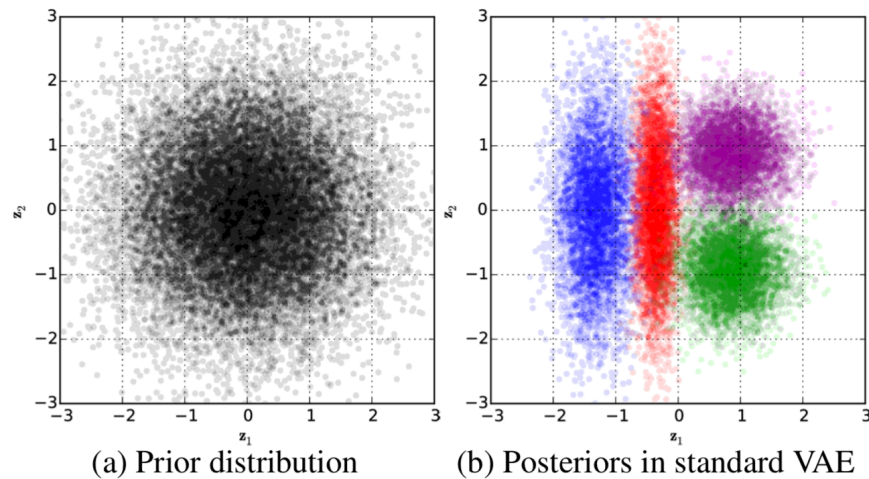
Optimal VAE Prior

$$D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z}) \parallel p(\mathbf{z})) = 0 \Leftrightarrow p(\mathbf{z}) = q_{\text{agg},\phi}(\mathbf{z}) = \frac{1}{n} \sum_{i=1}^n q_{\phi}(\mathbf{z}|\mathbf{x}_i).$$

Thus, the optimal prior distribution $p(\mathbf{z})$ is the aggregated variational posterior $q_{\text{agg},\phi}(\mathbf{z})$.

Recap of Previous Lecture

- **Prior mismatch:** The unimodal encoder $q_\phi(\mathbf{z}|\mathbf{x})$ yields $q_{\text{agg},\phi}(\mathbf{z})$ that often does not match the Gaussian prior $p(\mathbf{z})$.
- **Blurriness from averaging:** With Gaussian decoder, ELBO minimization gives $\mu^*(\mathbf{z}) = \mathbb{E}_{q_\phi(\mathbf{x}|\mathbf{z})}[\mathbf{x}]$. If distinct inputs $\mathbf{x} \neq \mathbf{x}'$ map to *overlapping* latent regions, the decoder averages over unrelated data.



Recap of Previous Lecture

Assumptions

- Let $c \sim \text{Cat}(\boldsymbol{\pi})$, where

$$\boldsymbol{\pi} = (\pi_1, \dots, \pi_K), \quad \pi_k = P(c = k), \quad \sum_{k=1}^K \pi_k = 1.$$

- Suppose the VAE adopts a discrete latent variable c with prior $p(c) = \text{Uniform}\{1, \dots, K\}$.

ELBO

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q_{\phi}(c|\mathbf{x})} \log p_{\theta}(\mathbf{x}|c) - D_{\text{KL}}(q_{\phi}(c|\mathbf{x}) \parallel p(c)) \rightarrow \max_{\phi, \theta}.$$

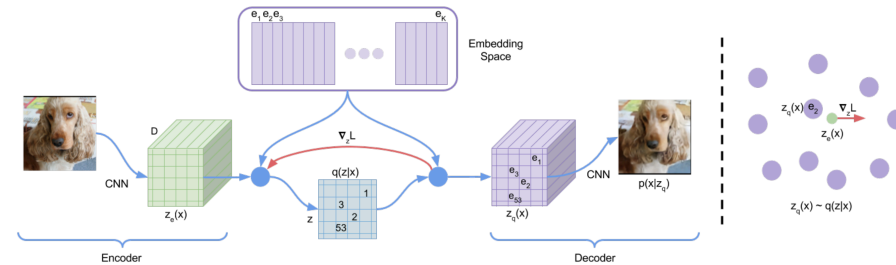
$$D_{\text{KL}}(q_{\phi}(c|\mathbf{x}) \parallel p(c)) = -H(q_{\phi}(c|\mathbf{x})) + \log K.$$

Quantized Representation

Define the codebook (dictionary) space $\{\mathbf{e}_k\}_{k=1}^K$ with $\mathbf{e}_k \in \mathbb{R}^L$ and K the number of codebook entries.

$$\mathbf{z}_q = \mathbf{q}(\mathbf{z}) = \mathbf{e}_{k^*}, \quad \text{where } k^* = \arg \min_k \|\mathbf{z} - \mathbf{e}_k\|.$$

Recap of Previous Lecture



Deterministic Variational Posterior

$$q_\phi(c = k^* | \mathbf{x}) = \begin{cases} 1, & \text{for } k^* = \arg \min_k \|\mathbf{z}_e - \mathbf{e}_k\|; \\ 0, & \text{otherwise.} \end{cases}$$

ELBO

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q_\phi(c|\mathbf{x})} \log p_\theta(\mathbf{x} | \mathbf{e}_c) - \log K = \log p_\theta(\mathbf{x} | \mathbf{z}_q) - \log K.$$

Straight-Through Gradient Estimator

$$\frac{\partial \log p(\mathbf{x} | \mathbf{z}_q, \boldsymbol{\theta})}{\partial \phi} = \frac{\partial \log p_\theta(\mathbf{x} | \mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_q}{\partial \phi} \approx \frac{\partial \log p_\theta(\mathbf{x} | \mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_e}{\partial \phi}$$

Recap of Previous Lecture

Likelihood-Free Learning

- Likelihood isn't always a suitable metric for evaluating generative models.
- Sometimes, the likelihood function can't even be computed exactly.

Imagine we have two sets of samples:

- $\{\mathbf{x}_i\}_{i=1}^{n_1} \sim p_{\text{data}}(\mathbf{x})$ — real samples;
- $\{\mathbf{x}_i\}_{i=1}^{n_2} \sim p_{\theta}(\mathbf{x})$ — generated (fake) samples.

$$p(y = 1|\mathbf{x}) = P(\mathbf{x} \sim p_{\text{data}}(\mathbf{x})); \quad p(y = 0|\mathbf{x}) = P(\mathbf{x} \sim p_{\theta}(\mathbf{x}))$$

Assumption

The generative model $p_{\theta}(\mathbf{x})$ matches $p_{\text{data}}(\mathbf{x})$ if a discriminative model $p(y|\mathbf{x})$ can't distinguish between them — that is, if $p(y = 1|\mathbf{x}) = 0.5$ for every $\mathbf{x}$.

- **Generator:** a generative model $\mathbf{x} = \mathbf{G}_{\theta}(\mathbf{z})$ that produces more realistic samples.
- **Discriminator:** a classifier $D_{\phi}(\mathbf{x}) \in [0, 1]$ distinguishing real from generated samples.

Recap of Previous Lecture

Cross-Entropy for Discriminative Model

$$\max_{p(y|\mathbf{x})} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log p(y = 1|\mathbf{x}) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log p(y = 0|\mathbf{x}) \right]$$

- **Discriminator:** A classifier $p_{\phi}(y = 1|\mathbf{x}) = D_{\phi}(\mathbf{x}) \in [0, 1]$, distinguishing real and generated samples. The discriminator aims to **maximize** cross-entropy.
- **Generator:** The generative model $\mathbf{x} = \mathbf{G}_{\theta}(\mathbf{z})$, $\mathbf{z} \sim p(\mathbf{z})$, seeks to fool the discriminator. The generator aims to **minimize** cross-entropy.

GAN Objective

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log(1 - D(\mathbf{x})) \right]$$

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D(\mathbf{G}(\mathbf{z}))) \right]$$

Outline

01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

03 Wasserstein GAN (WGAN)

04 Evaluation of Likelihood-Free Models

Frechet Inception Distance (FID) · Precision-Recall · CLIP Score · Human Eval

Outline

01 Generative Adversarial Networks (GAN)

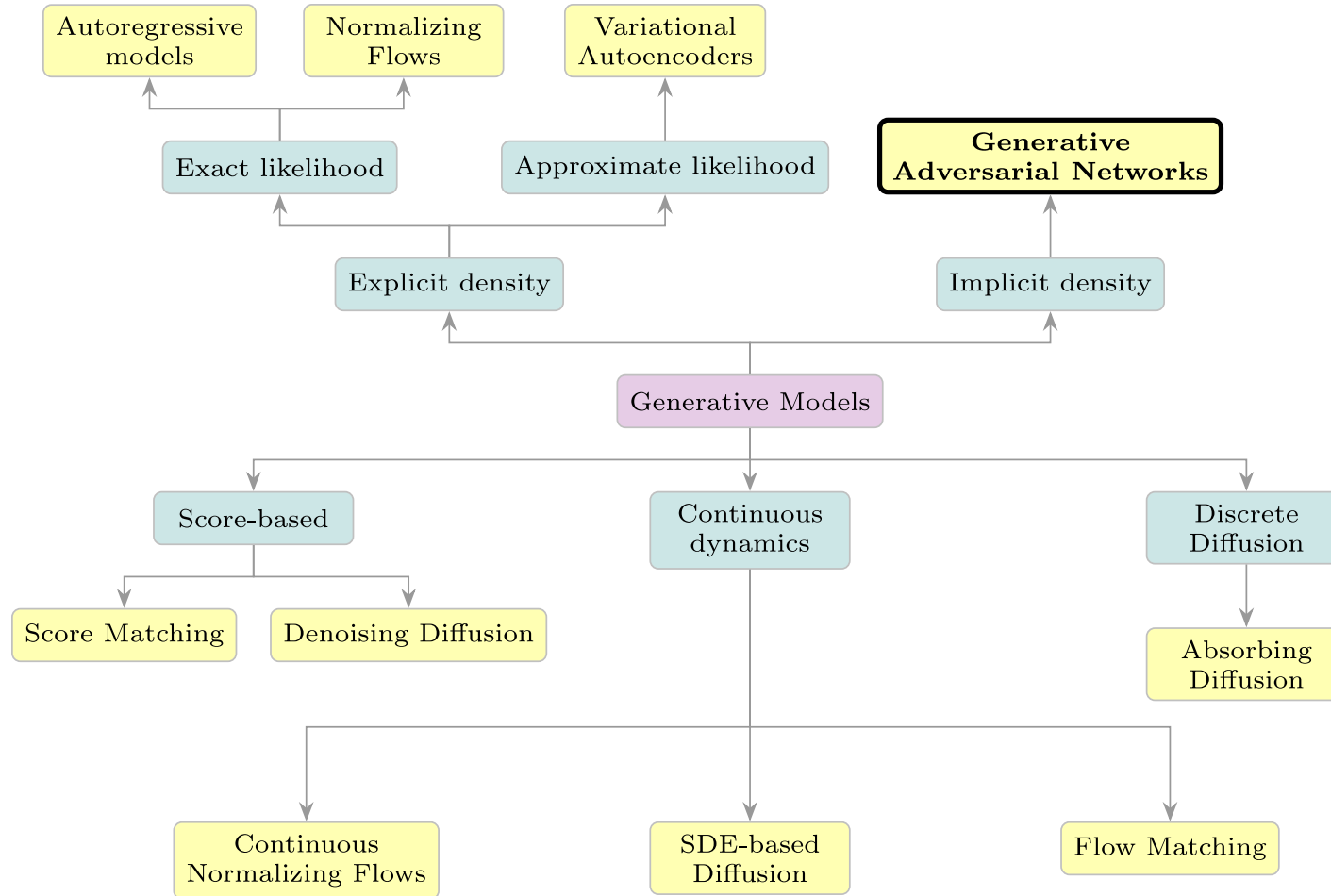
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Generative Models Taxonomy



GAN Optimality

Theorem

The minimax game

$$\min_G \max_D \underbrace{\left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D(\mathbf{G}(\mathbf{z}))) \right]}_{V(G,D)}$$

achieves its global optimum when $p_{\text{data}}(\mathbf{x}) = p_{\theta}(\mathbf{x})$, and $D^*(\mathbf{x}) = 0.5$.

Proof (Fixed G)

$$\begin{aligned} V(G, D) &= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log(1 - D(\mathbf{x})) \\ &= \int \underbrace{[p_{\text{data}}(\mathbf{x}) \log D(\mathbf{x}) + p_{\theta}(\mathbf{x}) \log(1 - D(\mathbf{x}))]}_{y(D)} d\mathbf{x} \\ \frac{dy(D)}{dD} &= \frac{p_{\text{data}}(\mathbf{x})}{D(\mathbf{x})} - \frac{p_{\theta}(\mathbf{x})}{1 - D(\mathbf{x})} = 0 \quad \Rightarrow \quad D^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\theta}(\mathbf{x})} \end{aligned}$$

GAN Optimality

Proof Continued (Fixed $D = D^*$)

$$\begin{aligned}
 V(G, D^*) &= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log \left(\frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\theta}(\mathbf{x})} \right) \\
 &\quad + \mathbb{E}_{p_{\theta}(\mathbf{x})} \log \left(\frac{p_{\theta}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\theta}(\mathbf{x})} \right) \\
 &= D_{\text{KL}} \left(p_{\text{data}}(\mathbf{x}) \parallel \frac{p_{\text{data}}(\mathbf{x}) + p_{\theta}(\mathbf{x})}{2} \right) + D_{\text{KL}} \left(p_{\theta}(\mathbf{x}) \parallel \frac{p_{\text{data}}(\mathbf{x}) + p_{\theta}(\mathbf{x})}{2} \right) - 2 \log 2 \\
 &= 2 D_{\text{JS}}(p_{\text{data}}(\mathbf{x}) \parallel p_{\theta}(\mathbf{x})) - 2 \log 2.
 \end{aligned}$$

Jensen-Shannon Divergence (Symmetric KL Divergence)

$$D_{\text{JS}}(p_{\text{data}}(\mathbf{x}) \parallel p_{\theta}(\mathbf{x})) = \frac{1}{2} [D_{\text{KL}}(p_{\text{data}}(\mathbf{x}) \parallel \star) + D_{\text{KL}}(p_{\theta}(\mathbf{x}) \parallel \star)]$$

This can be regarded as a proper distance metric!

$$V(G^*, D^*) = -2 \log 2, \quad p_{\text{data}}(\mathbf{x}) = p_{\theta}(\mathbf{x}), \quad D^*(\mathbf{x}) = 0.5.$$

GAN Optimality

Theorem

The following minimax game

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D(\mathbf{G}(\mathbf{z}))) \right]$$

achieves its global optimum precisely when $p_{\text{data}}(\mathbf{x}) = p_{\theta}(\mathbf{x})$, and $D^*(\mathbf{x}) = 0.5$.

Expectations

If the generator can express **any** function and the discriminator is **optimal** at every step, the generator **will converge** to the target distribution.

Reality

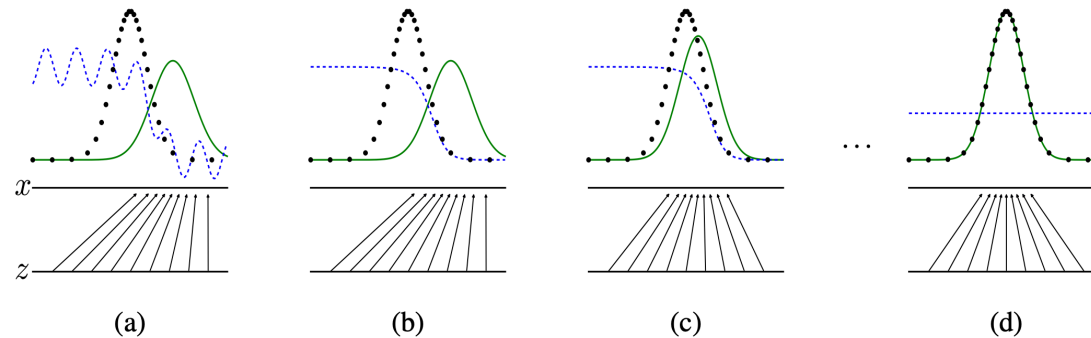
- Generator updates are performed in parameter space, and the discriminator is often imperfectly optimized.
- Generator and discriminator losses typically oscillate during GAN training.

GAN Training

Assume both generator and discriminator are parametric models: $D_\phi(\mathbf{x})$ and $\mathbf{G}_\theta(\mathbf{z})$.

Objective

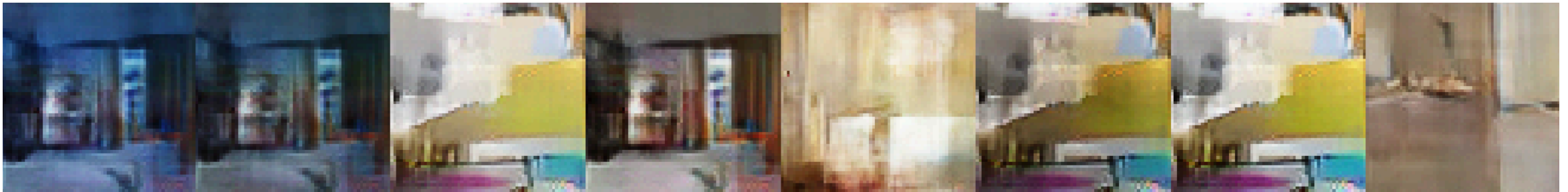
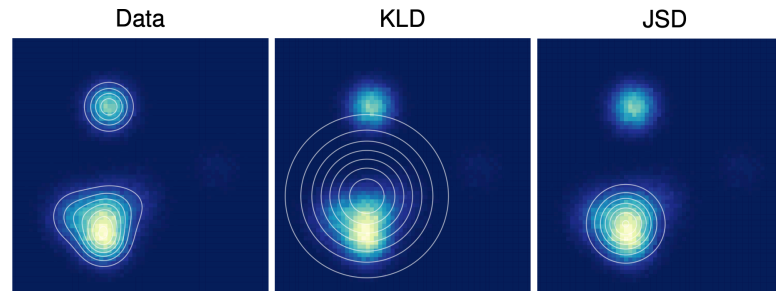
$$\min_{\theta} \max_{\phi} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_\phi(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_\phi(\mathbf{G}_\theta(\mathbf{z}))) \right]$$



- $\mathbf{z} \sim p(\mathbf{z})$ is a latent variable.
- $p_\theta(\mathbf{x}|\mathbf{z}) = \delta(\mathbf{x} - \mathbf{G}_\theta(\mathbf{z}))$ serves as a deterministic decoder (like normalizing flows).
- There is no encoder present.

Mode Collapse

Mode collapse refers to the phenomenon where the generator in a GAN produces only one or a few different modes of the distribution.



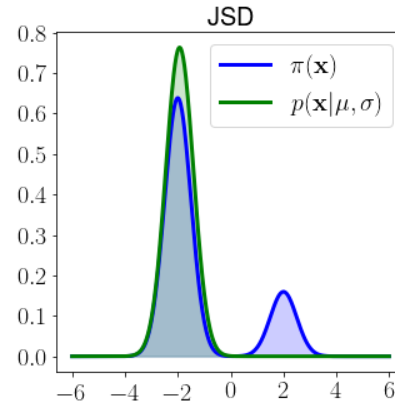
Numerous methods have been proposed to tackle mode collapse: changing architectures, adding regularization terms, injecting noise.

Jensen-Shannon vs Kullback-Leibler Divergences

- $p_{\text{data}}(\mathbf{x})$ is a fixed mixture of two Gaussians.
- $p(\mathbf{x}|\mu, \sigma) = \mathcal{N}(\mu, \sigma^2)$.

Mode Covering vs. Mode Seeking

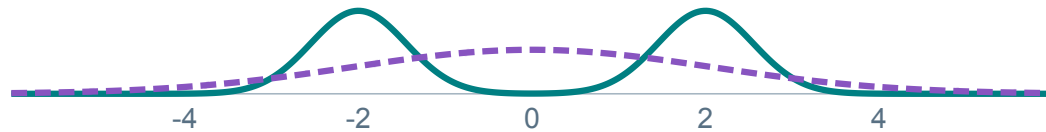
$$D_{\text{KL}}(\pi \parallel p) = \int \pi(\mathbf{x}) \log \frac{\pi(\mathbf{x})}{p(\mathbf{x})} d\mathbf{x}, \quad D_{\text{KL}}(p \parallel \pi) = \int p(\mathbf{x}) \log \frac{p(\mathbf{x})}{\pi(\mathbf{x})} d\mathbf{x}$$
$$D_{\text{JS}}(\pi \parallel p) = \frac{1}{2} \left[D_{\text{KL}}\left(\pi(\mathbf{x}) \parallel \frac{\pi(\mathbf{x}) + p(\mathbf{x})}{2}\right) + D_{\text{KL}}\left(p(\mathbf{x}) \parallel \frac{\pi(\mathbf{x}) + p(\mathbf{x})}{2}\right) \right]$$



Mode Covering vs. Mode Seeking

$$p_{\text{data}}(x) = \frac{1}{2}\mathcal{N}(x \mid -2, 0.55^2) + \frac{1}{2}\mathcal{N}(x \mid 2, 0.55^2), \quad p_{\theta}(x) = \mathcal{N}(x \mid \mu, \sigma^2).$$

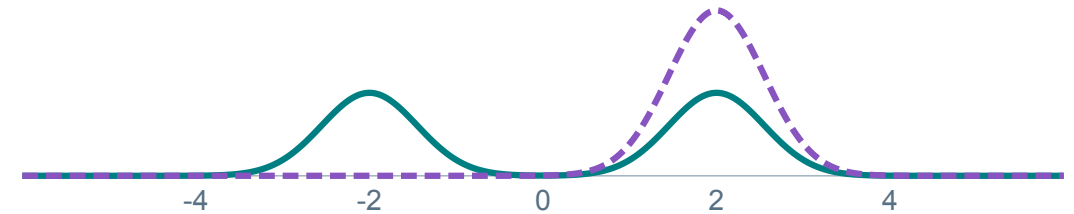
Forward KL optimum



$\mu = 0.00, \sigma = 2.07$

Forward KL 0.635 · Reverse KL 1.623

Reverse KL minimum (right mode)



$\mu = 2.00, \sigma = 0.55$

Forward KL 12.445 · Reverse KL 0.693

— Target: two modes

- - - - - Model: one Gaussian

Forward KL

Missing a target mode is expensive.

Reverse KL

Putting model mass in the low-density gap is expensive.

This difference comes from restricting the model to one Gaussian. If it could match the target exactly, both KLs would be zero.

Outline

01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

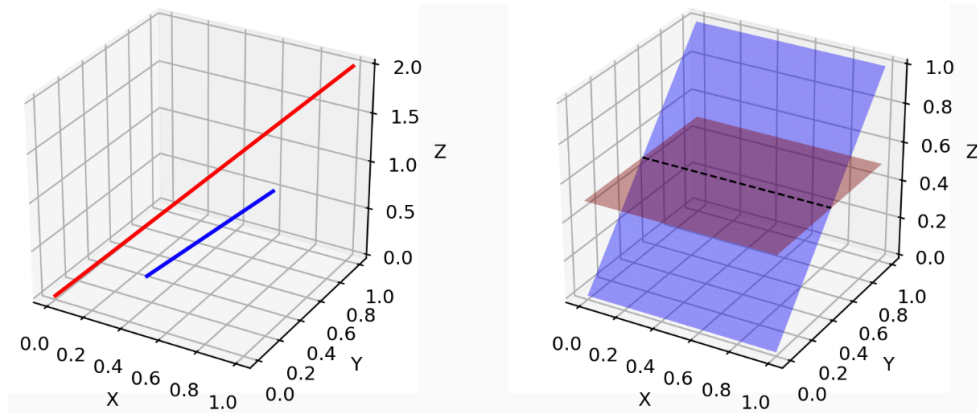
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Theoretical Results

- The dimensionality of $\mathbf{z}$ is less than that of $\mathbf{x}$, so $p_{\theta}(\mathbf{x})$ with $\mathbf{x} = \mathbf{G}_{\theta}(\mathbf{z})$ lives on a low-dimensional manifold.
- The true data distribution $p_{\text{data}}(\mathbf{x})$ is also supported on a low-dimensional manifold.



- If $p_{\text{data}}(\mathbf{x})$ and $p_{\theta}(\mathbf{x})$ are disjoint, a smooth optimal discriminator can exist!

For such low-dimensional, disjoint manifolds:

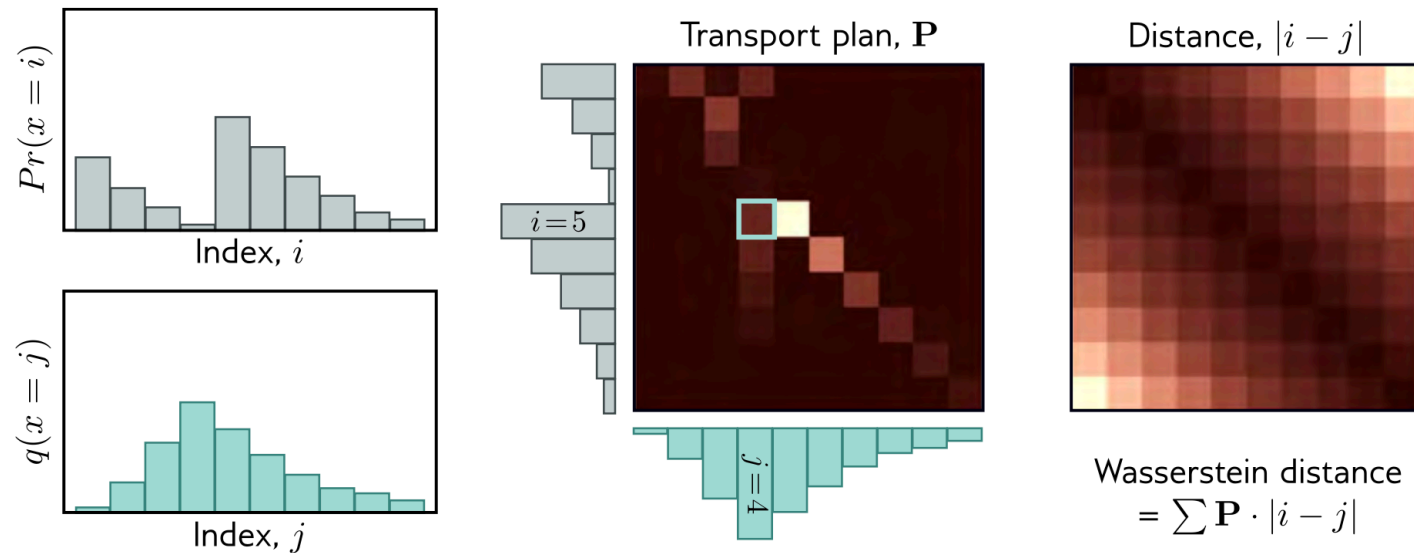
$$D_{\text{KL}}(p_{\text{data}} \parallel p_{\theta}) = D_{\text{KL}}(p_{\theta} \parallel p_{\text{data}}) = \infty, \quad D_{\text{JS}}(p_{\text{data}} \parallel p_{\theta}) = \log 2$$

Wasserstein Distance (Discrete)

Also known as the **Earth Mover's Distance**.

Optimal Transport Formulation

The minimum cost of moving and transforming a pile of “dirt” shaped like one probability distribution to match another.



Wasserstein Distance (Continuous)

$$\begin{aligned}
 W(\pi||p) &= \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| \\
 &= \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2
 \end{aligned}$$

- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the transport plan: the amount of “dirt” assigned from $\mathbf{x}_1$ to $\mathbf{x}_2$.

$$\int \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 = p(\mathbf{x}_2); \quad \int \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_2 = \pi(\mathbf{x}_1).$$

- $\Gamma(\pi, p)$ denotes the set of all joint distributions $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ with marginals π and p .
- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the mass, $\|\mathbf{x}_1 - \mathbf{x}_2\|$ is the distance.

Wasserstein Metric

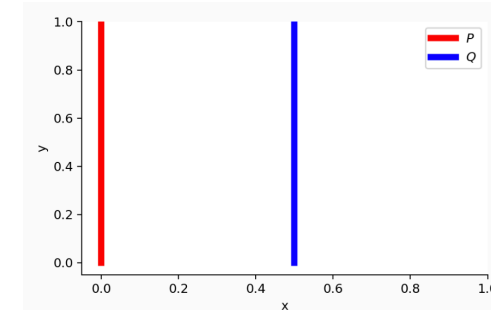
$$W_s(\pi, p) = \inf_{\gamma \in \Gamma(\pi, p)} \left(\mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\|^s \right)^{1/s}$$

In our setting, $W(\pi||p) = W_1(\pi, p)$, which is the transport cost formulation.

Wasserstein Distance vs KL vs JSD

Consider two-dimensional distributions:

$$p_{\text{data}}(x, y) = (0, U[0, 1]), \quad p_{\theta}(x, y) = (\theta, U[0, 1])$$



$\theta = 0$: Both distributions are identical.

$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = D_{\text{KL}}(p_{\theta} \| p_{\text{data}}) = D_{\text{JS}}(p_{\theta} \| p_{\text{data}}) = W(p_{\text{data}} \| p_{\theta}) = 0$$

$\theta \neq 0$:

$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = \int_{U[0,1]} 1 \log \frac{1}{0} dy = \infty = D_{\text{KL}}(p_{\theta} \| p_{\text{data}})$$

$$D_{\text{JS}}(p_{\text{data}} \| p_{\theta}) = \frac{1}{2} \left(\int_{U[0,1]} 1 \log \frac{1}{1/2} dy + \int_{U[0,1]} 1 \log \frac{1}{1/2} dy \right) = \log 2$$

$$W(p_{\text{data}} \| p_{\theta}) = |\theta|$$

Wasserstein Distance vs KL vs JSD

Theorem 1

Let $G_\theta(\mathbf{z})$ be (almost) any feedforward neural network, and $p(\mathbf{z})$ a prior over $\mathbf{z}$ such that $\mathbb{E}_{p(\mathbf{z})} \|\mathbf{z}\| < \infty$. Then $W(p_{\text{data}} \| p_\theta)$ is continuous everywhere and differentiable almost everywhere.

Theorem 2

Let π be a distribution on a compact space $\mathcal{X}$ and let $\{p_t\}_{t=1}^\infty$ be a sequence of distributions on $\mathcal{X}$.

$$D_{\text{KL}}(\pi \| p_t) \rightarrow 0 \quad (\text{or } D_{\text{KL}}(p_t \| \pi) \rightarrow 0) \quad (1)$$

$$D_{\text{JS}}(\pi \| p_t) \rightarrow 0 \quad (2)$$

$$W(\pi \| p_t) \rightarrow 0 \quad (3)$$

As $t \rightarrow \infty$, $(1) \Rightarrow (2)$, and $(2) \Rightarrow (3)$. That is, convergence in Wasserstein distance is a weaker condition than convergence in JSD, which in turn is weaker than convergence in KL.

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Wasserstein GAN

Wasserstein Distance

$$\begin{aligned} W(\pi \| p) &= \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| \\ &= \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2 \end{aligned}$$

The infimum over all possible $\gamma \in \Gamma(\pi, p)$ is computationally intractable.

Theorem (Kantorovich-Rubinstein Duality)

$$W(\pi \| p) = \frac{1}{K} \max_{\|f\|_L \leq K} \left[\mathbb{E}_{\pi(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p(\mathbf{x})} f(\mathbf{x}) \right]$$

where $f : \mathbb{R}^m \rightarrow \mathbb{R}$ is K -Lipschitz ($\|f\|_L \leq K$):

$$|f(\mathbf{x}_1) - f(\mathbf{x}_2)| \leq K \|\mathbf{x}_1 - \mathbf{x}_2\|, \quad \forall \mathbf{x}_1, \mathbf{x}_2 \in \mathcal{X}.$$

We can thus estimate $W(\pi \| p)$ using only samples and a function f .

Wasserstein GAN

Theorem (Kantorovich-Rubinstein Duality)

$$W(p_{\text{data}} \| p_{\theta}) = \frac{1}{K} \max_{\|f\|_L \leq K} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\theta}(\mathbf{x})} f(\mathbf{x}) \right]$$

- We must ensure that f is K -Lipschitz continuous.
- Let $f_{\phi}(\mathbf{x})$ be a feedforward neural network parameterized by ϕ .
- If the weights ϕ are restricted to a compact set Φ , then f_{ϕ} is K -Lipschitz.
- Clamp weights within the box $\Phi = [-c, c]^d$ (e.g. $c = 0.01$) after each update.

$$\begin{aligned} K \cdot W(p_{\text{data}} \| p_{\theta}) &= \max_{\|f\|_L \leq K} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\theta}(\mathbf{x})} f(\mathbf{x}) \right] \\ &\geq \max_{\phi \in \Phi} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\phi}(\mathbf{x}) - \mathbb{E}_{p_{\theta}(\mathbf{x})} f_{\phi}(\mathbf{x}) \right] \end{aligned}$$

Wasserstein GAN

Standard GAN Objective

$$\min_{\theta} \max_{\phi} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\phi}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\phi}(\mathbf{G}_{\theta}(\mathbf{z})))$$

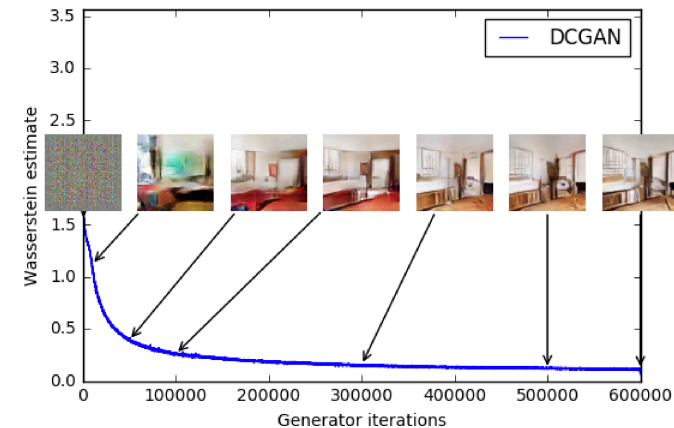
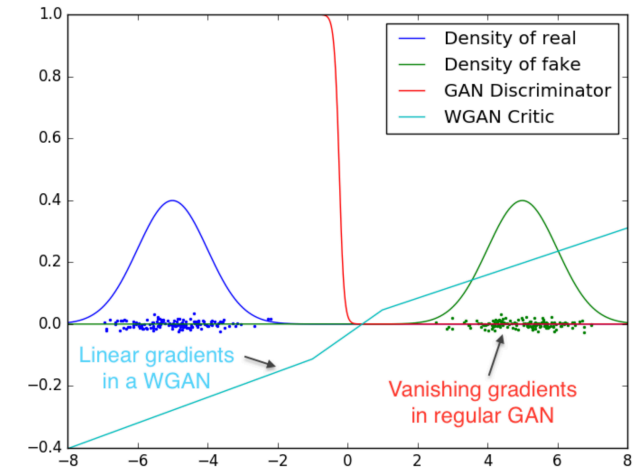
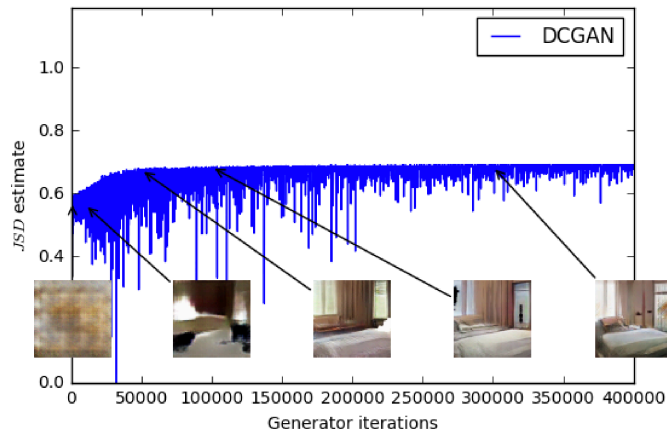
WGAN Objective

$$\min_{\theta} W(p_{\text{data}} || p_{\theta}) \approx \min_{\theta} \max_{\phi \in \Phi} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\phi}(\mathbf{x}) - \mathbb{E}_{p(\mathbf{z})} f_{\phi}(\mathbf{G}_{\theta}(\mathbf{z})) \right]$$

- The discriminator D is replaced by function f : in WGAN, it is known as the **critic**, which is *not* a classifier.
- “*Weight clipping is a clearly terrible way to enforce a Lipschitz constraint.*”
 - If c is large, optimizing the critic is hard.
 - If c is small, gradients may vanish.

Wasserstein GAN

- WGAN provides nonzero gradients even if distributions' supports are disjoint.
- $D_{JS}(p_{\text{data}} || p_{\theta})$ is poorly correlated with sample quality and remains near its maximum value $\log 2 \approx 0.69$.
- $W(p_{\text{data}} || p_{\theta})$ is tightly correlated with quality.



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Evaluation of Likelihood-Free Models

Likelihood-Based Models

- **Train:** fit the model.
- **Validation:** tune hyperparameters.
- **Test:** assess generalization by reporting likelihood.

Not all models have tractable likelihoods (VAE: compare ELBO values; GAN: ???).

Desirable Properties for Samples

Sharpness



Low sharpness

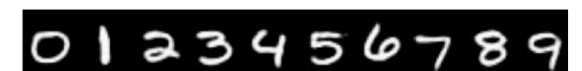


High sharpness

Diversity



Low diversity



High diversity

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Frechet Inception Distance (FID) · Precision-Recall · CLIP Score · Human Eval

Wasserstein Metric

$$W_s(\pi||p) = \inf_{\gamma \in \Gamma(\pi, p)} \left(\mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\|^s \right)^{1/s}$$

Wasserstein GAN (Optimal Transport)

$$\begin{aligned} W(\pi||p) &= \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\| \\ &= \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 d\mathbf{x}_2 \end{aligned}$$

Theorem

If $\pi(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\pi, \boldsymbol{\Sigma}_\pi)$, $p(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p)$, then

$$W_2^2(\pi||p) = \|\boldsymbol{\mu}_\pi - \boldsymbol{\mu}_p\|^2 + \text{tr} \left[\boldsymbol{\Sigma}_\pi + \boldsymbol{\Sigma}_p - 2 \left(\boldsymbol{\Sigma}_\pi^{1/2} \boldsymbol{\Sigma}_p \boldsymbol{\Sigma}_\pi^{1/2} \right)^{1/2} \right]$$

Frechet Inception Distance

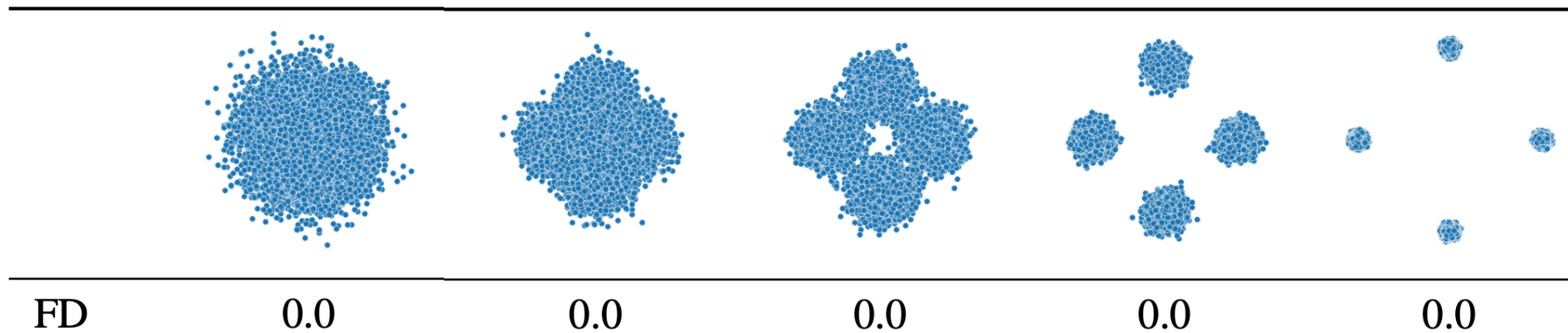
$$\text{FID}(p_{\text{data}}, p_{\boldsymbol{\theta}}) = W_2^2(p_{\text{data}}||p_{\boldsymbol{\theta}})$$

Frechet Inception Distance (FID)

$$\text{FID}(p_{\text{data}}, p_{\theta}) = \|\boldsymbol{\mu}_{\text{data}} - \boldsymbol{\mu}_{\theta}\|^2 + \text{tr} \left[\boldsymbol{\Sigma}_{\text{data}} + \boldsymbol{\Sigma}_{\theta} - 2 \left(\boldsymbol{\Sigma}_{\text{data}}^{1/2} \boldsymbol{\Sigma}_{\theta} \boldsymbol{\Sigma}_{\text{data}}^{1/2} \right)^{1/2} \right]$$

- FID is computed in the latent space $\mathbf{z}$.
- We use a pretrained image embedder to get latent representations $\mathbf{z} = \mathbf{f}(\mathbf{x})$.
- $\boldsymbol{\mu}_{\text{data}}$, $\boldsymbol{\Sigma}_{\text{data}}$ and $\boldsymbol{\mu}_{\theta}$, $\boldsymbol{\Sigma}_{\theta}$ are statistics of latent representations for samples from $p_{\text{data}}(\mathbf{x})$ and $p_{\theta}(\mathbf{x})$.

$\text{FID}(p(\mathbf{x}), \mathcal{N}(0, \mathbf{I}))$



Frechet Inception Distance (FID)

$$\text{FID}(p_{\text{data}}, p_{\theta}) = \|\mu_{\text{data}} - \mu_{\theta}\|^2 + \text{tr} \left[\Sigma_{\text{data}} + \Sigma_{\theta} - 2 \left(\Sigma_{\text{data}}^{1/2} \Sigma_{\theta} \Sigma_{\text{data}}^{1/2} \right)^{1/2} \right]$$

Drawbacks

- Depends on the pretrained classification network.
- Uses the normality assumption.
- May not correlate with human evaluation.

Model	Model-A	Model-B
FID	21.40	18.42
FID _∞	20.16	17.19
Human rater preference	92.5%	6.9%

Outline

01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

03 Wasserstein GAN (WGAN)

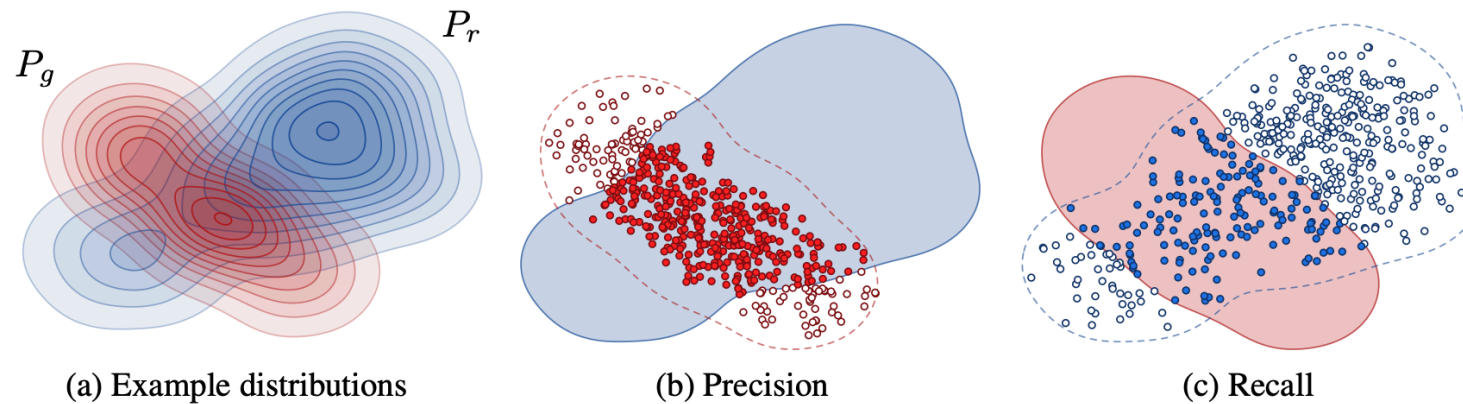
04 Evaluation of Likelihood-Free Models

Frechet Inception Distance (FID) · **Precision-Recall** · CLIP Score · Human Eval

Precision-Recall

Desirable Properties for Samples

- **Sharpness:** generated samples should possess high visual quality.
- **Diversity:** their variation should match that in the training data.



- **Precision** denotes the fraction of generated images that look realistic.
- **Recall** measures how well the generator covers the training data manifold.

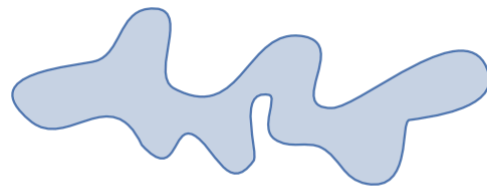
Precision-Recall

- $\mathcal{S}_{\text{data}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\text{data}}(\mathbf{x})$ — real samples;
- $\mathcal{S}_{\theta} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\theta}(\mathbf{x})$ — generated samples.

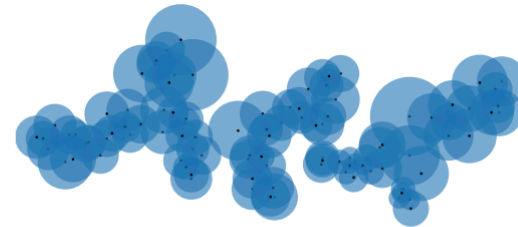
Define a binary function:

$$\mathbb{I}(\mathbf{x}, \mathcal{S}) = \begin{cases} 1, & \text{if } \exists \mathbf{x}' \in \mathcal{S} : \|\mathbf{x} - \mathbf{x}'\|_2 \leq \|\mathbf{x}' - \text{NN}_k(\mathbf{x}', \mathcal{S})\|_2; \\ 0, & \text{otherwise.} \end{cases}$$

$$\text{Pr}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\theta}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\theta}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\text{data}}); \quad \text{Rec}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\theta}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\text{data}}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\theta}).$$



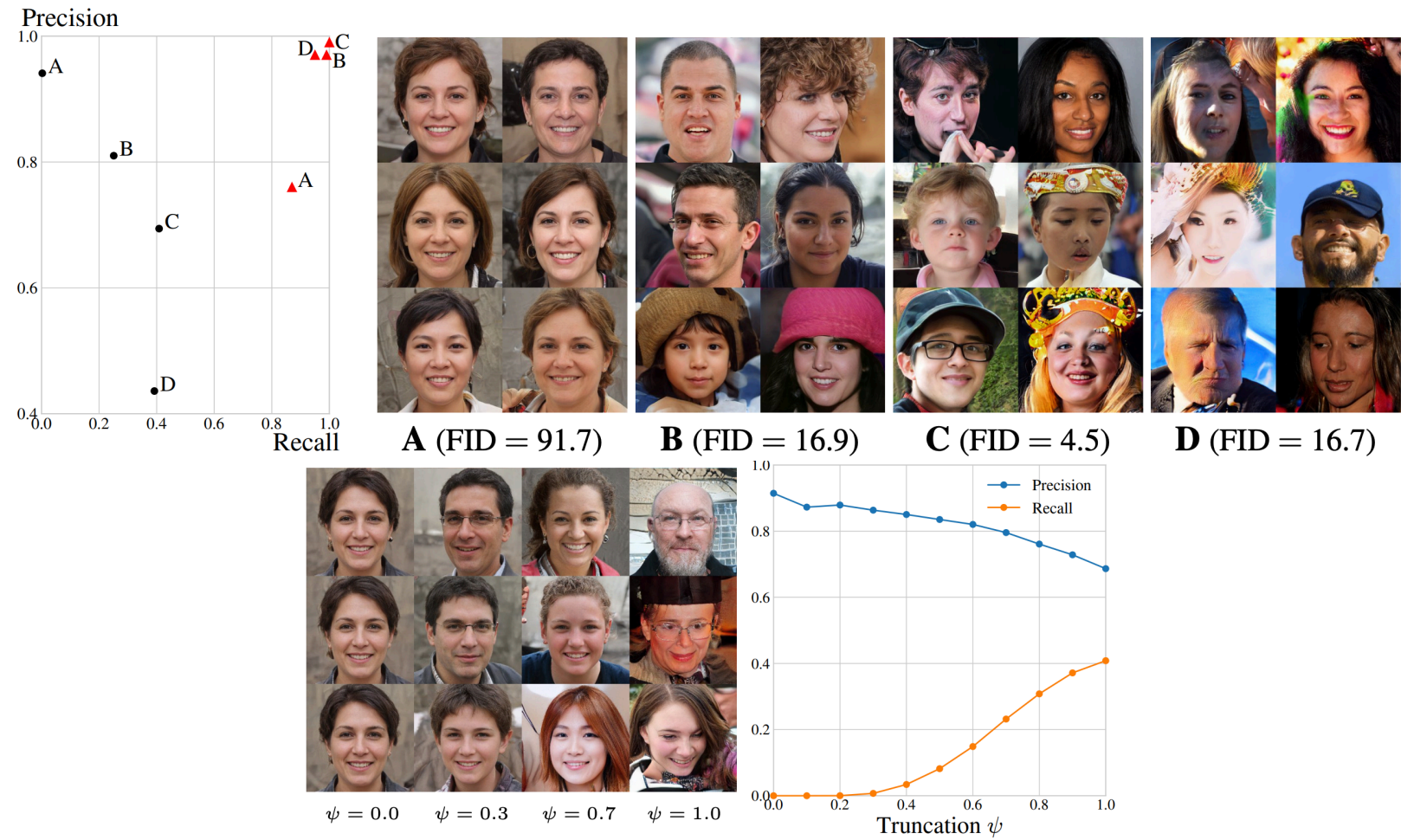
(a) True manifold



(b) Approx. manifold

Embed the samples using a pretrained network (as in FID).

Precision-Recall



Outline

01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

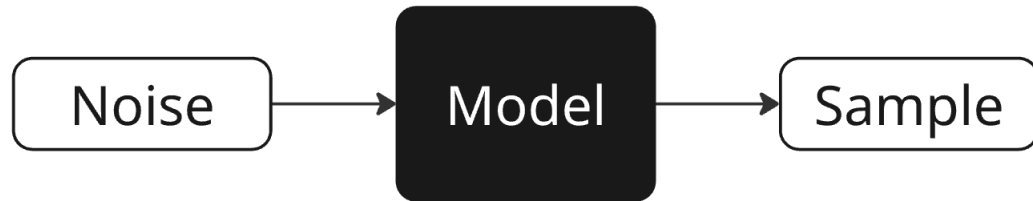
03 Wasserstein GAN (WGAN)

04 Evaluation of Likelihood-Free Models

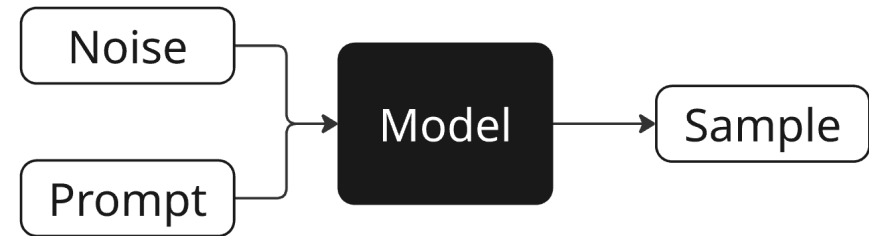
Frechet Inception Distance (FID) · Precision-Recall · **CLIP Score** · Human Eval

CLIP Score

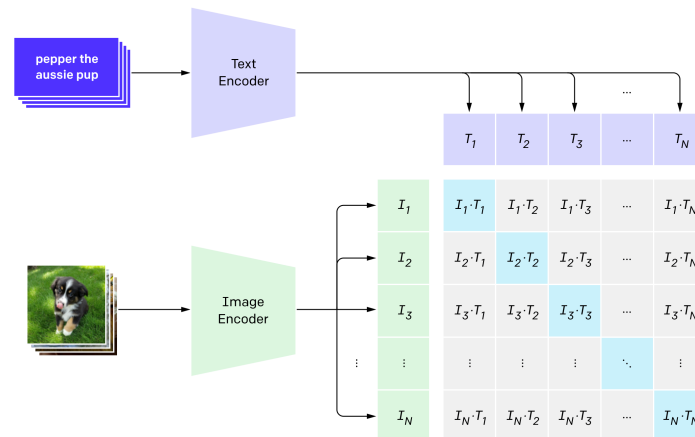
Unconditional Model



Conditional Model



We need a way to measure not only the quality of the generated image, but also how well it's aligned with the prompt.



Outline

01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

03 Wasserstein GAN (WGAN)

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Frechet Inception Distance (FID) · Precision-Recall · CLIP Score · **Human Eval**

Human Evaluation

- No automated metric is perfect.
- The best way to evaluate generative models is by human assessment.
- It's important to assess various properties.

Аспект	Yandex ART 2.0	Mj 6.1	Mj 6	Ideogram	Recraft	Google Imagen3	Dall-E 3	FLUX	SBER Kandi3.1
Релевантность	0,59	0,58	0,63	0,45	0,51	0,50	0,50	0,54	0,75
Эстетика	0,49	0,55	0,55	0,51	0,51	0,61	0,61	0,54	0,59
Комплексность	0,44	0,73	0,70	0,68	0,76	0,75	0,75	0,71	0,74
Дефектность	0,69	0,57	0,68	0,55	0,59	0,63	0,63	0,50	0,75
Предпочтение	0,66	0,60	0,69	0,49	0,54	0,63	0,63	0,51	0,84

Summary

- GANs, in theory, optimize the Jensen-Shannon divergence.
- Wasserstein distance works in the case of disjoint data and model distributions (unlike the KL and JS divergences).
- Wasserstein GAN uses Kantorovich-Rubinstein duality to enable Monte Carlo estimation of the Wasserstein distance. It enforces the Lipschitz condition on the critic through weight clipping.
- FID is the most popular metric for evaluating implicit generative models.
- Precision-recall allows for choosing a model that balances sample quality and diversity.
- The CLIP score is widely used to measure text-to-image alignment.
- The gold standard for evaluating generated image quality is human assessment.